1.01 Indices

Chapter 1 · Original slide 4 · 1: Index Laws

base

3^5

exponent or index (plural: indices)

!

$a^m \times a^n = a^{m+n}$
$a^m \div a^n = a^{m-n}$
$(a^m)^n = a^{mn}$
$(ab)^n = a^n b^n$

power

Some also refer to the 5 as the ‘power’. But this would lead to unfortunate ambiguous phrases like “when we multiply two powers together, we add the powers” instead of “when we multiply two powers together, we add the indices”.

Simplify $\left(a^3\right)^2 \times 2a^2$

Simplify $\left(4x^3y\right)^3$

Simplify $2x^2\left(3+5x\right) - x\left(4-x^2\right)$

Simplify $\frac{x^3-2x}{3x^2}$

Tip: A common student error is to get the sign wrong of $+x^3$

Tip: While $\frac{a+b}{c}$ can be split into $\frac{a}{c} + \frac{b}{c}$, a common student error is to think that $\frac{a}{b+c} = \frac{a}{b} + \frac{a}{c}$

Chapter 1 · Original slide 5 · Test Your Understanding

1

Simplify $\left(\frac{2a^5}{a^2}\right)^2 \times 3a$

2

Simplify $\frac{2x+x^5}{4x^3}$

3

Expand and simplify
$2x\left(3-x^2\right) - 4x^3\left(3-x\right)$

Note: This is using $(ab)^n = a^n b^n$ law backwards.

4

Simplify $2^x \times 3^x$

1.01 Indices

Chapter 1 · Original slide 6 · Exercise 1A – Pearson Pure Y1 page 4

2 Expand and simplify if possible:

o $x(3x^2 - 2x + 5)$
p $7y^2(2 - 5y + 3y^2)$
q $-2y^2(5 - 7y + 3y^2)$
r $7(x - 2) + 3(x + 4) - 6(x - 2)$
s $5x - 3(4 - 2x) + 6$
t $3x^2 - x(3 - 4x) + 7$
u $4x(x + 3) - 2x(3x - 7)$
v $3x^2(2x + 1) - 5x^2(3x - 4)$

d $\frac{8x^3+5x}{2x}$ **e** $\frac{7x^7+5x^2}{5x}$ **f** $\frac{9x^5-5x^3}{3x}$

3 Simplify these fractions:

Chapter 1 · Original slide 7 · Exercise 1A – Pearson Pure Y1 page 4

Extension (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

1

[MAT 2006 1A] Which of the following numbers is largest?

$$\left((2^3)^2\right)^3, \quad (2^3)^{(2^3)}, \quad 2^{((3^2)^3)}, \quad 2^{(3^{(2^3)})}$$

2

[MAT 2012 1B] Let $N = 2^k \times 4^m \times 8^n$ where k,m,n are positive whole numbers. Then N will definitely be a square number whenever:

k is even;

$k + n$ is odd;

k is odd but $m + n$ is even;

$k + n$ is even.

1.01 Indices

Chapter 1 · Original slide 17 · 4: Negative and Fractional Indices

$a^0 = 1$
$a^{\frac{1}{m}} = \sqrt[m]{a}$
$a^{\frac{n}{m}} = \left(\sqrt[m]{a}\right)^n$
$a^{-m} = \frac{1}{a^m}$

Note: $\sqrt{9}$ only means the positive square root of 9, i.e. 3 not -3.

Otherwise, what would be the point of the $\pm$ in the quadratic formula before the $\sqrt{b^2 - 4ac}$?

Prove that $x^{\frac{1}{2}} = \sqrt{x}$

Evaluate $27^{-\frac{1}{3}}$

Evaluate $32^{\frac{2}{5}}$

Simplify $\left(\frac{1}{9}x^6y\right)^{\frac{1}{2}}$

Evaluate $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$

If $b = \frac{1}{9}a^2$, determine $3b^{-2}$ in the form kb^n where k, n are constants.

Chapter 1 · Original slide 18 · Writing a surd using indices

If $9\sqrt{3} = 3^k$, determine the value of k.

The key here is to write everything as powers with a consistent base, in this case, 3.

Note: In algebra we like to avoid mixed numbers. So don't write $3^{2\frac{1}{2}}$

Chapter 1 · Original slide 19 · Further Examples

[Edexcel IGCSE May14-4H] Given that $\left(2^{\frac{1}{2}}\right)^n = \frac{2^x}{8^y}$

Express n in terms of x and y.

Write 8 as a power of 2 (putting a bracket around it), for consistency of base with the other powers.

[Edexcel IAL C12 Jan 2019 Q2c] Given that $y = 2^x$, express the following in terms of y.

$\frac{1}{4^{2x-3}}$

Write your expression in its simplest form.

Chapter 1 · Original slide 20 · Test Your Understanding

Given $2^x \times 4^y = \frac{1}{2\sqrt{2}}$ express y as a function of x. (3)

Chapter 1 · Original slide 21 · Exercise 1D

4 a Find the value of $81^{\frac{1}{4}}$. **b** Simplify $x(2x^{-\frac{1}{3}})^4$.

5 Given that $y = \frac{1}{8}x^3$ express each of the following in the form kx^n, where k and n are constants.

a $y^{\frac{1}{3}}$ **b** $\frac{1}{2}y^{-2}$

Challenge